

Module 3: Data Link Layer and Wireless Networks



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Data-Link Layer

- Data Link Control (DLC)
- Multiple Access Protocols (MAC)
- Link-Layer Addressing
- Ethernet Protocol
- Connecting Devices

Wireless Networks

- Wireless LANs
- Mobile IP

Hands-On

- Datalink Provider Interface
- SOCK_PACKET
- PF_PACKET

- Mobile and personal computers are increasingly popular
- Traditional IP assumes hosts remain fixed to one network
- Mobile IP extends IP to support mobility
- Allows Internet connectivity from different locations
- Key challenge: maintaining communication while moving

Mobile IP: Addressing Problem

- Main challenge in Mobile IP is addressing
- Original IP assumes hosts are stationary
- Host is permanently attached to one network
- Mobility violates this assumption
- New addressing mechanisms are required

Stationary Hosts and IP Addressing

- Traditional IP designed for stationary hosts
- Router uses IP address to route datagrams
- IP address consists of:

- Prefix
- Suffix



- Prefix identifies the network
- Example:
- 10.3.4.24/8 belongs to network 10.0.0.0/8

Why Traditional Addressing Works

Key Observation

IP addresses are designed to work with stationary hosts because part of the address defines the network to which the host is attached.

- Address tied to a specific network
- Valid only while attached to that network
- If network changes, address becomes invalid
- Routers use prefix to identify destination network
- Works perfectly for stationary hosts

Mobile Hosts and the Need for Change

- Mobile hosts move between networks
- Existing IP structure becomes inadequate
- Addressing mechanism must support mobility
- Several solutions have been proposed

Solution 1: Changing the Address

- Mobile host obtains a new IP address
- DHCP assigns address in the new network
- New address associates host with the new network

Drawbacks

- Configuration files need updates
- DNS tables require modification
- Host may need rebooting
- Active communications may be interrupted

Problems During Communication

- ① Address changes disrupt active connections
- ② Client and server require fixed IP addresses
- ③ Port numbers must remain unchanged
- ④ Data exchange may be interrupted

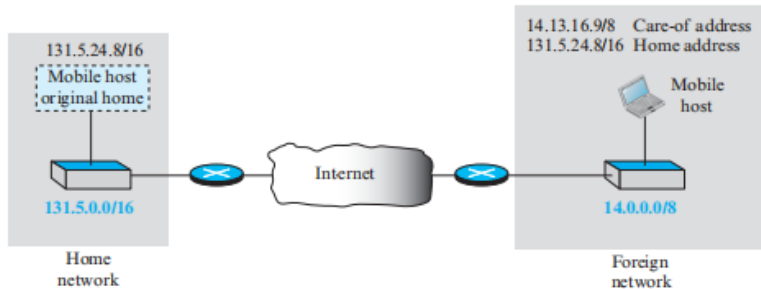
Major Problem

Client and server IP addresses must remain constant for the duration of the connection.

Solution 2: Two Addresses

- More practical solution
- Mobile host maintains:
 - Home Address
 - Care-of Address
- Supports mobility without changing identity
- Enables seamless communication

Figure 6.52 *Home address and care-of address*



Home Address

- Permanent address of mobile host
- Associated with the home network
- Remains unchanged regardless of location
- Used as the host's long-term identity

Characteristics

Permanent Stable Globally Recognizable

Care-of Address

- Temporary address assigned in a foreign network
- Represents current location of the mobile host
- Changes whenever the host moves
- Associated with the foreign network

Characteristics

Temporary Dynamic Location Dependent

Home Network vs Foreign Network

Home Network

- Permanent network of the mobile host
- Associated with home address

Foreign Network

- Network currently visited by the host
- Associated with care-of address

INSERT FIGURE HERE

Mobile Host Moving from Home Network to Foreign Network

Obtaining the Care-of Address

- Mobile host receives care-of address after entering a foreign network
- Address assignment occurs during:
 - Agent Discovery Phase
 - Registration Phase
- Care-of address identifies current location
- Enables packet forwarding to the mobile host

Mobile IP Addressing Summary

Feature	Home Address	Care-of Address
Nature	Permanent	Temporary
Changes	No	Yes
Associated With	Home Network	Foreign Network
Purpose	Identity	Current Location

Key Idea

Mobile IP uses two addresses:

- Home Address (Permanent)
- Care-of Address (Temporary)

to maintain connectivity while supporting mobility.

- To make the change of address transparent to the rest of the Internet, Mobile IP requires:
 - Home Agent
 - Foreign Agent
- Home agent is associated with the home network
- Foreign agent is associated with the foreign network
- Both are usually implemented on routers

INSERT FIGURE 6.53 HERE

Home Agent and Foreign Agent

Agents in Mobile IP

- Home and foreign agents are shown as routers
- Their agent functionality is performed at the application layer
- They perform dual roles:
 - Router
 - Host
- Support communication with mobile hosts

Important

Home agents and foreign agents are routers, but their agent functionality operates at the application layer.

Home Agent

- Usually a router attached to the home network
- Acts on behalf of the mobile host
- Receives packets sent to the mobile host
- Forwards packets toward the foreign agent

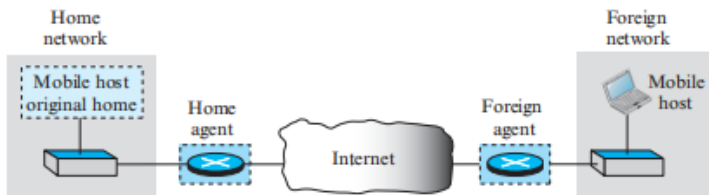
Function

Remote Host → Home Agent → Foreign Agent

Role of the Home Agent

- Maintains knowledge of the mobile host's location
- Receives packets addressed to the home address
- Redirects packets toward the current location
- Provides transparency to communicating hosts

Figure 6.53 *Home agent and foreign agent*



Foreign Agent

- Usually a router attached to the foreign network
- Receives packets from the home agent
- Delivers packets to the mobile host
- Represents the mobile host in the foreign network

Function

Home Agent → Foreign Agent → Mobile Host

Mobile Host as a Foreign Agent

- Mobile host can also act as a foreign agent
- Mobile host and foreign agent become the same entity
- Eliminates dependence on an external foreign agent

Key Idea

A mobile host can act as its own foreign agent.

Requirements for Self-Acting Mobile Host

- Must obtain a care-of address independently
- DHCP can be used to obtain the care-of address
- Must contain additional Mobile IP software
- Must communicate directly with the home agent

Dual Addressing Requirement

- Mobile host must maintain:
 - Home Address
 - Care-of Address
- Both addresses coexist simultaneously
- Dual addressing must be transparent to applications
- Application programs should not be aware of mobility

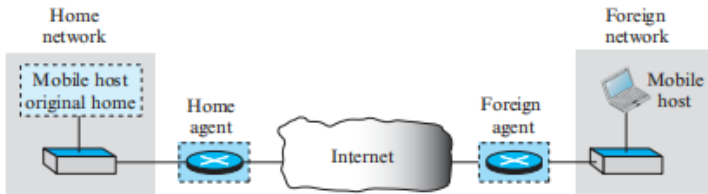
Transparency

Mobility support should not require modification of application programs.

- Occurs when the mobile host acts as its own foreign agent
- Mobile host and foreign agent become identical
- Care-of address is known as:

Collocated Care-of Address

Figure 6.53 *Home agent and foreign agent*



Advantages of Collocated Care-of Address

- Mobile host can move to any network
- No need to locate an available foreign agent
- Greater flexibility and mobility
- Simplifies deployment in foreign networks

Advantage

Movement is independent of foreign-agent availability.

Disadvantages of Collocated Care-of Address

- Mobile host requires additional software
- Increased processing burden on the host
- Host must perform foreign-agent functions
- Additional management complexity

Disadvantage

The mobile host must act as its own foreign agent.

Summary: Agents in Mobile IP

Agent	Function
Home Agent	Receives packets for mobile host and forwards them to foreign network
Foreign Agent	Receives forwarded packets and delivers them to the mobile host
Mobile Host as Foreign Agent	Uses a collocated care-of address and performs foreign-agent functions itself

Key Point

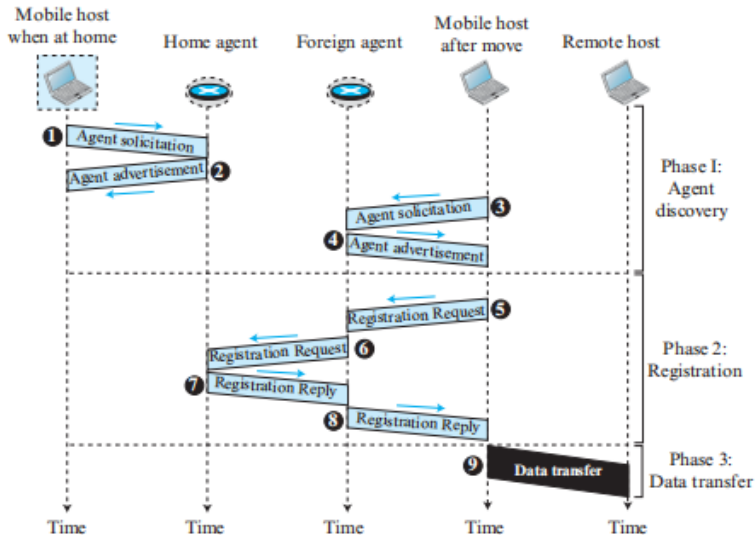
When the mobile host and the foreign agent are the same, the care-of address is called a **Collocated Care-of Address**.

Three Phases of Mobile Communication

- Communication in Mobile IP occurs in three phases:
 - ① Agent Discovery
 - ② Registration
 - ③ Data Transfer
- First phase involves:
 - Mobile Host
 - Home Agent
 - Foreign Agent
- Third phase additionally involves the Remote Host



Figure 6.54 *Remote host and mobile host communication*



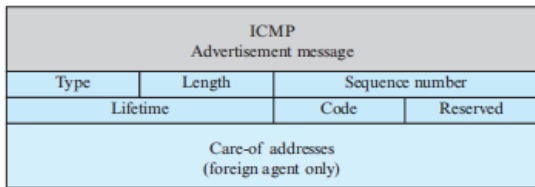
Phase 1: Agent Discovery

- First phase of Mobile IP communication
- Consists of two subphases:
 - ① Discover Home Agent
 - ② Discover Foreign Agent
- Mobile host learns:
 - Foreign Agent Address
 - Care-of Address
- Uses:
 - Agent Advertisement
 - Agent Solicitation

Agent Advertisement

- Router advertises its presence using ICMP Router Advertisement
- If router acts as an agent:
 - Agent Advertisement is appended
- Mobile IP does not define a new packet type
- Uses existing ICMP Router Advertisement packet

Figure 6.55 *Agent advertisement*



Agent Advertisement Fields

- Type (8 bits)
 - Value = 16
- Length (8 bits)
 - Total extension length
- Sequence Number (16 bits)
 - Detects lost messages
- Lifetime
 - Duration agent accepts requests

Agent Advertisement Fields (Contd.)

- Code Field (8 bits)
 - Contains capability flags
- Care-of Address List
 - Available care-of addresses
 - Mobile host selects one address
 - Used only by Foreign Agent

Agent Advertisement Code Bits

Bit	Meaning
0	Registration required; no collocated care-of address
1	Agent busy; registration not accepted
2	Agent acts as Home Agent
3	Agent acts as Foreign Agent
4	Minimal Encapsulation supported
5	GRE supported
6	Header Compression supported
7	Unused

Agent Solicitation

- Used when no Agent Advertisement is received
- Initiated by Mobile Host
- Uses ICMP Router Solicitation message
- Requests assistance from available agents

Important

Mobile IP does not define a new packet type for solicitation. It uses ICMP Router Solicitation.

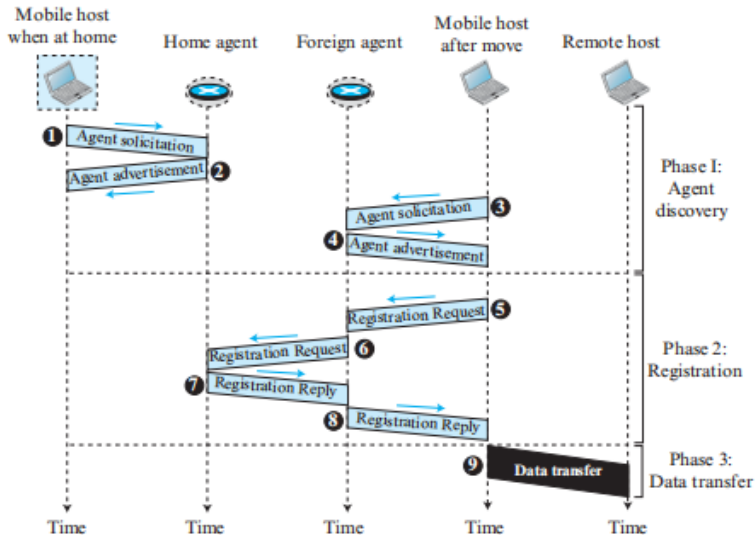
Phase 2: Registration

- Performed after discovering a Foreign Agent
- Mobile Host must register
- Four aspects:
 - 1 Register with Foreign Agent
 - 2 Register with Home Agent
 - 3 Renew registration when expired
 - 4 Deregister when returning home

Registration Messages

- Registration Request
- Registration Reply
- Request sent by Mobile Host
- Reply generated by Home Agent

Figure 6.54 *Remote host and mobile host communication*



Registration Request

- Sent from Mobile Host to Foreign Agent
- Registers Care-of Address
- Announces:
 - Home Address
 - Home Agent Address
- Foreign Agent relays request to Home Agent

Figure 6.56 *Registration request format*

Type	Flag	Lifetime
Home address		
Home agent address		
Care-of address		
Identification		
Extensions ...		

Registration Request Fields

- Type = 1
- Flag Field
- Lifetime
- Home Address
- Home Agent Address
- Care-of Address
- Identification Field
- Authentication Extensions

Registration Request Flag Bits

Bit	Meaning
0	Retain prior care-of address
1	Tunnel broadcast messages
2	Collocated care-of address used
3	Request minimal encapsulation
4	Request GRE encapsulation
5	Request header compression
6-7	Reserved

Registration Reply

- Sent by Home Agent
- Relayed through Foreign Agent
- Confirms or denies registration
- Similar to Registration Request
- Type = 3
- Flag field replaced by Code field
- Care-of address field removed

Figure 6.57 *Registration reply format*

Type	Code	Lifetime
Home address		
Home agent address		
Identification		
Extensions ...		

Encapsulation of Registration Messages

- Registration messages are carried inside UDP
- Agent uses well-known port:

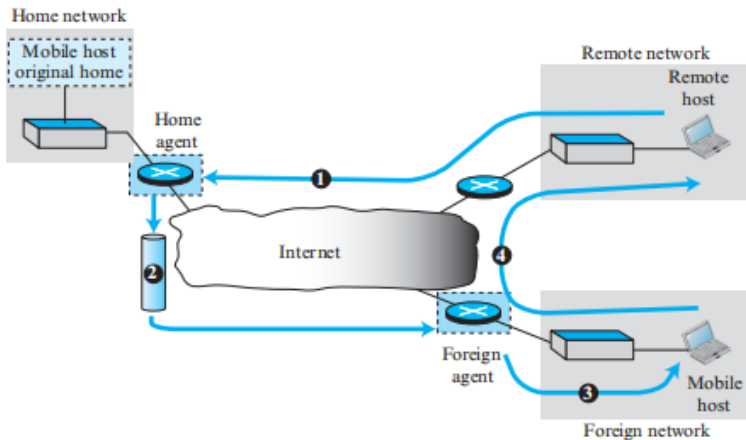
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- Mobile Host uses an ephemeral port

Phase 3: Data Transfer

- Begins after:
 - Agent Discovery
 - Registration
- Mobile Host communicates with Remote Host
- Home Agent and Foreign Agent participate

Figure 6.58 *Data transfer*



Path 1: Remote Host to Home Agent

- Remote Host sends packet normally
- Source = Remote Host Address
- Destination = Home Address of Mobile Host
- Home Agent intercepts packet
- Uses Proxy ARP

Path 1

Remote Host → Home Agent

Path 2: Home Agent to Foreign Agent

- Home Agent forwards packet
- Uses tunneling mechanism
- Original IP packet is encapsulated
- New packet:
 - Source = Home Agent
 - Destination = Foreign Agent

Path 2

Home Agent → Foreign Agent

Path 3: Foreign Agent to Mobile Host

- Foreign Agent removes outer packet
- Consults registration table
- Finds corresponding Care-of Address
- Delivers packet to Mobile Host

Path 3

Foreign Agent → Mobile Host

Path 4: Mobile Host to Remote Host

- Mobile Host sends reply normally
- Source = Home Address
- Destination = Remote Host Address
- Packet originates from foreign network
- Home Address remains unchanged

Path 4

Mobile Host → Remote Host

Transparency in Mobile IP

- Remote Host is unaware of mobility
- Packets sent to Home Address
- Replies originate from Home Address
- Movement remains invisible
- Communication continues seamlessly

Key Point

The mobility of the Mobile Host is completely transparent to the rest of the Internet.

Summary of Mobile IP Operation

- ① Agent Discovery
 - Advertisement
 - Solicitation
- ② Registration
 - Request
 - Reply
- ③ Data Transfer
 - Remote Host → Home Agent
 - Home Agent → Foreign Agent
 - Foreign Agent → Mobile Host
 - Mobile Host → Remote Host



6.3.4 Inefficiency in Mobile IP

- Mobile IP communication may be inefficient
- Packets may travel longer paths than necessary
- Inefficiency can be:
 - ① Double Crossing (2X Routing)
 - ② Triangle Routing (Dog-leg Routing)

Key Idea

Packets may follow indirect routes because communication is anchored at the Home Agent.

Types of Mobile IP Inefficiency

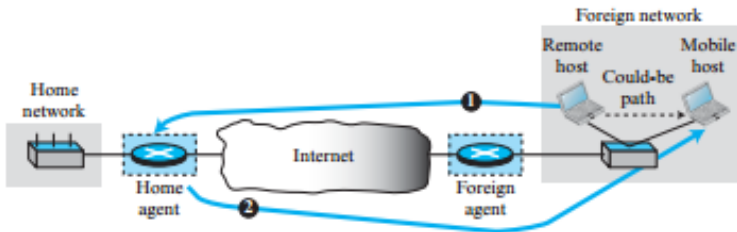
Type	Severity
Double Crossing	Severe
Triangle Routing	Moderate

- Both inefficiencies occur during packet delivery
- Mainly affect packets sent from:
 - Remote Host
 - Mobile Host

Double Crossing (2X Routing)

- Most severe inefficiency in Mobile IP
- Occurs when:
 - Remote Host and Mobile Host are in the same network/site
 - Mobile Host has moved away from its home network
- Packet delivery still passes through the Home Agent
- Results in unnecessary Internet traversal

Figure 6.59 *Double crossing*



Double Crossing: Packet Flow

Mobile Host → Remote Host

- Communication is local
- No inefficiency
- Packet follows direct route

Remote Host → Mobile Host

- Packet sent to Home Address
- Intercepted by Home Agent
- Forwarded back to Mobile Host
- Packet crosses Internet twice

Why Double Crossing is Serious

- Most communication is local
- Based on the Principle of Locality
- Nearby hosts communicate frequently
- Double crossing converts local traffic into long-distance traffic
- Causes:
 - Increased delay
 - Wasted bandwidth
 - Reduced efficiency



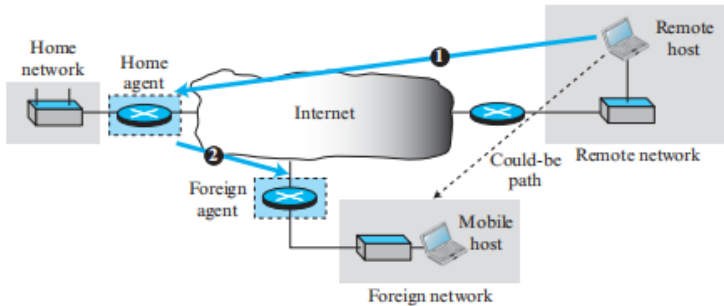
Principle of Locality

Computers usually communicate more frequently with nearby computers than distant computers.

Triangle Routing (Dog-leg Routing)

- Less severe than Double Crossing
- Occurs when:
 - Remote Host and Mobile Host are in different networks
- Communication path becomes triangular
- Packet takes two sides of a triangle instead of one

Figure 6.60 *Triangle routing*



Triangle Routing: Packet Flow

Mobile Host → Remote Host

- Direct communication
- No inefficiency

Remote Host → Mobile Host

- Packet sent to Home Agent
- Home Agent forwards packet
- Packet finally reaches Mobile Host
- Indirect route is followed

Triangle Routing Path



- Packet traverses two links
- Direct communication would require only one link
- Creates additional delay
- Consumes extra network resources

Double Crossing vs Triangle Routing

Feature	Double Crossing	Triangle Routing
Severity	High	Moderate
Communication Site	Same Site	Different Sites
Internet Traversal	Twice	Once Extra
Delay	Very High	Moderate
Bandwidth Waste	High	Moderate

Solution to Routing Inefficiency

- Use Address Binding
- Bind:
 - Home Address
 - Care-of Address
- Remote Host stores the binding information
- Future packets can be sent directly
- Avoids unnecessary Home Agent forwarding

Binding Update Mechanism

- ① Home Agent receives first packet
- ② Packet forwarded to Foreign Agent
- ③ Home Agent sends Binding Update
- ④ Remote Host learns Care-of Address
- ⑤ Future packets sent directly

Result

Triangle routing can be avoided after the first packet.

Binding Cache

- Remote Host stores binding information
- Stored in a Binding Cache
- Maps:

Home Address ↔ Care-of Address
- Subsequent packets use cached information

Problem with Binding Cache

- Mobile Host may move again
- Care-of Address changes
- Cached information becomes outdated
- Packets may be sent to wrong location

Problem

Binding cache entries can become stale.

Maintaining Correct Bindings

- Home Agent monitors location changes
- When Mobile Host moves:
 - Care-of Address changes
 - Previous binding becomes invalid
- Home Agent sends warning/update packet
- Remote Host updates its cache

Summary: Inefficiency in Mobile IP

- Mobile IP introduces routing inefficiencies
- Two major cases:
 - ① Double Crossing (2X Routing)
 - ② Triangle Routing
- Both arise due to Home Agent forwarding
- Binding updates reduce inefficiency
- Cache maintenance is necessary when hosts move

Key Point

The trade-off for mobility is increased routing complexity and possible routing inefficiency.